Problem 7

Evaluate the integral:

$$I = \int \frac{x^2+3}{x^2-1}\,dx.$$

Solution

To solve the given integral, we first decompose the fraction and integrate each term separately.

Step 1: Decompose the fraction

The integrand can be rewritten by splitting $\frac{x^2+3}{x^2-1}$ as follows:

$$\frac{x^2+3}{x^2-1} = 1 + \frac{4}{x^2-1}.$$

Thus, the integral becomes:

$$I = \int \frac{x^2+3}{x^2-1}\,dx = \int 1\,dx + \int \frac{4}{x^2-1}\,dx.$$

Step 2: Integrate each term

1. **First term: $\int 1\,dx$:**

$$\int 1\,dx = x.$$

2. **Second term: $\int \frac{4}{x^2-1}\,dx$:** The term $\frac{1}{x^2-1}$ can be decomposed using partial fractions:

$$\frac{1}{x^2-1} = \frac{1}{(x-1)(x+1)} = \frac{1}{2}\left(\frac{1}{x-1} - \frac{1}{x+1}\right).$$

Therefore:

$$\frac{4}{x^2-1} = 2\left(\frac{1}{x-1} - \frac{1}{x+1}\right).$$

Substituting this back, we get:

$$\int \frac{4}{x^2-1}\,dx = \int 2\left(\frac{1}{x-1} - \frac{1}{x+1}\right)dx.$$

Now, integrate each term:

$$\int \frac{4}{x^2-1}\,dx = 2\ln|x-1| - 2\ln|x+1| = 2\ln\left|\frac{x-1}{x+1}\right|.$$

Step 3: Combine the results

Combining the results of the two terms, we have:

$$I = x + 2\ln\left|\frac{x-1}{x+1}\right| + C,$$

where C is the constant of integration.

Final Answer

The solution to the integral is:

$$I = x + 2\ln\left|\frac{x-1}{x+1}\right| + C.$$